Project Update Proposal

- Personal Data Exposure Checker
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Current System Overview

- Checks email and phone number exposure
- Uses multiple APIs (HIBP + DeHashed)
- Risk scoring system (in progress)
- Privacy-preserving lookup

Identified Issues

- Multiple APIs increase complexity
- Phone number support is limited and inconsistent
- Higher integration and maintenance overhead
- Harder to ensure privacy consistency

Cost Consideration (DeHashed API)

- Cost for credits : (30$)
  can cost more if we hit the credits limit
- Impact: increases cost, not ideal for academic prototype

Proposed Change

- Remove phone number exposure checking
- Add password exposure checking
- Use only one API: Have I Been Pwned (HIBP)

Why Password Instead of Phone Number?

- Passwords are directly linked to security risk
- HIBP provides reliable password breach database
- Supports k-anonymity (privacy-preserving)
- More relevant for user protection

Advantages of Using Only HIBP

- Simplifies system architecture
- Strong privacy design (no raw password sent)
- Well-documented and widely used
- Eliminates multiple API management
- Improves reliability and performance

Updated System Features

- Email exposure checking (HIBP)
- Password exposure checking (HIBP)
- Risk scoring system (frequency, recency, sensitivity)
- Security recommendations

Impact on Risk Scoring

- Password exposure = high sensitivity factor
- Improves accuracy of risk scoring
- Reflects real-world attack risks
- Enhances user understanding

Updated System Flow

- User inputs email or password
- Data processed securely (hashing)
- Query sent to HIBP API
- Response received
- Risk score calculated
- Results and recommendations displayed

Expected Benefits

- More focused and practical system
- Stronger cybersecurity relevance
- Improved privacy compliance
- Easier implementation and testing
- Better user value

Supervisor Feedback Request

- Is removing phone number acceptable?
- Is focusing on HIBP only appropriate?
- Suggestions for improving scoring system?

Thank You

- Thank you for your guidance